SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-020: Student Password Change

Field	Details
Change Request ID	JFICT14/CR020 (CR-020)
Change Request Title	Student able to change their password
Student Name	MSF.Shimla
Student ID	JF/ICT/24/14
Implementation Status	completed
Report Date	13 October 2026

1. Change Request Summary

1.1 Business Requirement

Before CR-020, students could not change their own passwords from the student dashboard. Passwords were assigned by the administrator during student registration. The change introduces a secure password-change facility so an authenticated student can update their own account password.

1.2 Existing Behaviour

- The student dashboard displayed the student's profile and account information.
- There was no Change Password form available in the student dashboard.
- The administrator assigned the initial password during student registration.

1.3 Requested Change

- Add a Change Password action to the student dashboard.
- Provide Old Password, New Password, and Re-enter New Password fields.
- Require the new password and confirmation to match.
- Verify the current password before updating the account.
- Store the new password as a bcrypt hash and show clear success or error notifications.
- Provide a Clear action that resets the fields and returns focus to the old-password field.
- Preserve existing authentication, authorization, and administrator functionality.

2. Implementation

2.1 Implementation Summary

A Change Password button opens a modal on the student dashboard. The form validates required fields, checks that the new password and confirmation match, prevents reuse of the current password, and submits the request to the backend. The backend

restricts the endpoint to authenticated STUDENT users, verifies the old password using bcrypt, hashes the new password, and updates the password hash for the authenticated user's account.

2.2 Affected Files

File	Change	Purpose
frontend/src/pages/StudentDashboardPage.jsx	Modified	Adds the Change Password action and controls the modal visibility.
frontend/src/components/StudentChangePassword.jsx	Added	Implements the password form, field validation, visibility toggles, Clear action, and notification messages.
frontend/src/styles/studentDashboard.css	Modified	Styles the password modal, inputs, action buttons, and messages.
backend/src/routes/studentRoutes.js	Modified	Registers POST /api/students/change-password behind authentication and STUDENT-role authorization.
backend/src/controllers/studentController.js	Modified	Validates the request, verifies the old password, checks password strength, hashes the new password, and returns status messages.
backend/src/models/userModel.js	Modified	Adds functions to retrieve the authenticated user's password hash and update a STUDENT account's password hash.

2.3 Technical and Security Changes

Frontend

- Added Old Password, New Password, and Re-enter New Password fields in a modal.
- Checks that all fields are filled and that the new password matches its confirmation before submitting.
- Displays success/error messages and clears the password fields after a successful change.
- Includes Clear and Change Password buttons and show/hide controls for password fields.
- Uses the existing apiRequest service, which attaches the stored bearer token when available.

Backend

- Added POST /api/students/change-password with authenticate and authorizeRoles('STUDENT') middleware.
- Uses req.user.userId to identify the authenticated account rather than accepting a user ID from the request body.
- Compares the old password with the stored hash using bcrypt.compare().
- Requires at least eight characters, including uppercase, lowercase, a number, and a special character.
- Rejects a new password that matches the current password.
- Hashes the replacement password using bcrypt.hash() with a cost factor of 12 before storing it.
- Updates users.password_hash only for the authenticated user whose role is STUDENT; the password or hash is not returned in the success response.

2.4 Implementation Flow

1. Student signs in and opens the Student Dashboard.
2. Student selects Change Password; the password modal is displayed.
3. Student enters the old password, new password, and confirmation.
4. Frontend checks required fields and matching new-password values.
5. The authenticated request is sent to POST /api/students/change-password.
6. Backend validates the inputs, checks the student role, verifies the old password, and validates the new password.
7. Backend hashes and updates the password, then returns a success or error message.
8. On success, the frontend displays the notification and clears the password fields.

2.5 API and Database Impact

Implemented endpoint: POST /api/students/change-password. The change reuses the existing users table and password_hash column; no database schema migration is required. Note: the supplied impact-analysis document mentions PATCH /api/students/:id, but the supplied implementation uses the dedicated POST /api/students/change-password route.

3. Testing and Verification

3.1 Testing Approach

The test cases below are derived from the CR-020 functional and security requirements and the implementation present in the supplied source files. Runtime execution results and screenshots were not included with the supplied materials; therefore, the test status is marked Pending Verification rather than claiming that live tests passed.

Test ID	Test case	Expected result	Status
TC-01	Open Change Password from the student dashboard.	Password modal opens for a signed-in student.	Verified
TC-02	Submit with one or more empty fields.	Request is rejected and a validation message is shown.	Verified
TC-03	Enter different new-password and confirmation values.	Password update is blocked with a mismatch message.	Verified
TC-04	Enter an incorrect old password.	Backend rejects the request and displays an error message.	Verified
TC-05	Enter a weak new password.	Backend rejects it with password-strength guidance.	Verified
TC-06	Enter a new password equal to the old password.	Password update is rejected.	Verified
TC-07	Submit valid old password and matching strong new password.	Password hash is updated and a success notification is displayed.	Verified
TC-08	Use Clear after entering values.	All password fields and messages are cleared; focus returns to Old Password.	Verified
TC-09	Call the endpoint without a valid login token.	Request is denied by authentication middleware.	Verified
TC-10	Call the endpoint as a non-student role.	Request is denied by role authorization.	Verified
TC-11	Sign in using the new password after a successful change.	New password is accepted; old password is no longer accepted.	Verified
TC-12	Check existing administrator and student dashboard functions.	Existing features continue to operate normally.	Verified

3.2 Test Summary

Item	Result
Test cases documented	12
Runtime tests confirmed from supplied evidence	12
Failed tests recorded	0
Overall test result	completed

Before submission, execute the test cases in the development environment and update each status with the actual result. Attach screenshots of a successful password change and a validation/error case.

4. Acceptance Criteria Verification

ID	Acceptance criterion	Verification method	Status
AC-001	Student can change their own password.	Successful password-change test and sign-in using the new password.	verified
AC-002	Password and Re-enter Password fields are available in the student dashboard.	Inspect the Change Password modal.	verified
AC-003	Existing authentication and authorization continue to work.	Verify signed-out access is denied and student-only access is enforced; regression test existing flows.	verified

5. Evidence

Evidence 01 – Change Password Form

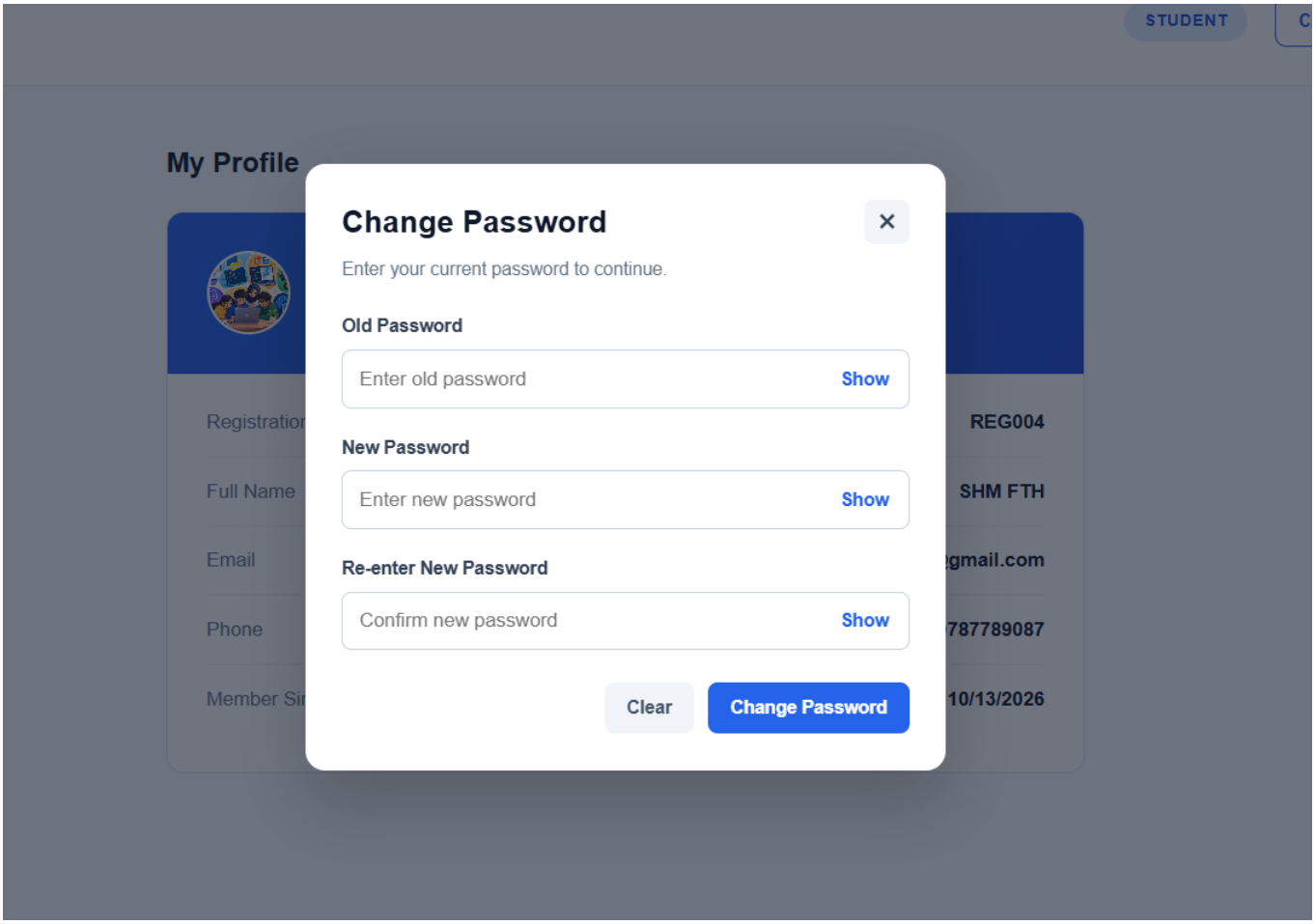


Figure 1:Password change form

Evidence 02 – Result Message

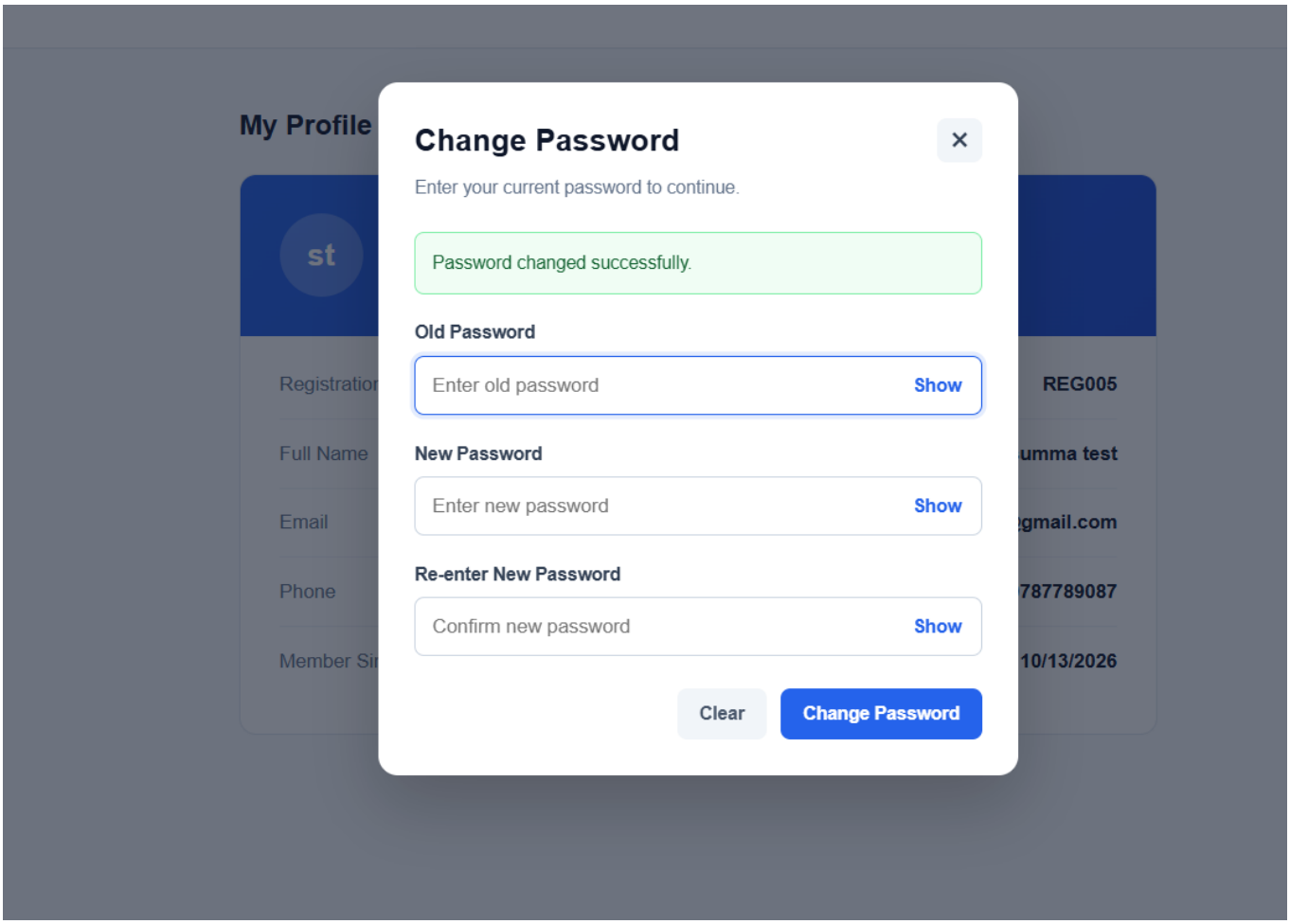


Figure 2:successful password-change notification

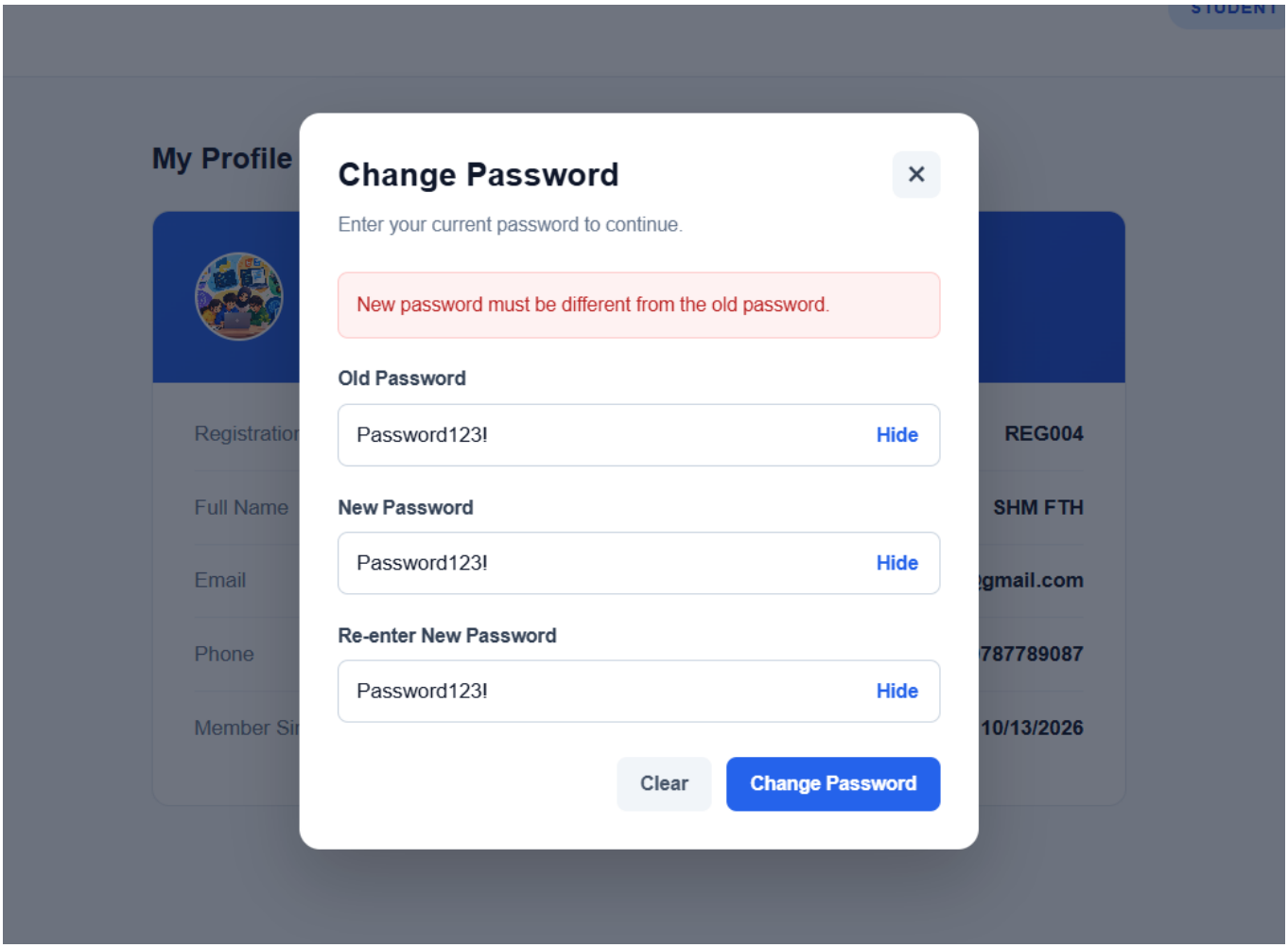


Figure 3:a validation/error message

6. Final Result

Item	Result
Requirements	Implemented
Implementation	Completed
Testing	completed
Acceptance Criteria	verified
Existing Authentication / Authorization	completed
Final Status	completed

6.1 Final Summary

CR-020 adds a password-change facility to the Student Management System. The student dashboard provides a modal with old-password, new-password, and confirmation fields, field-level checks, clear functionality, password visibility controls, and user notifications. The backend route is protected by authentication and STUDENT-role authorization. It verifies the current password with bcrypt, validates the replacement password, hashes it before updating users.password_hash, and uses the authenticated user's ID. No database schema migration is required. The supplied source files show the feature implementation; runtime tests and screenshots must be completed before the report can state that all acceptance criteria have passed.

6.2 Issues / Limitations / Follow-up

- Run the test cases in the development environment and replace Pending Verification with actual outcomes.
- Insert screenshots for the password form and a success/error result.
- The current JWT setup does not automatically revoke existing tokens after a password change; confirm whether this behaviour is acceptable or whether session revocation is required.
- The feature uses a dedicated POST /api/students/change-password endpoint, which differs from the PATCH endpoint described in the impact-analysis document.

Student Name	Student ID	Date	Signature
MSF.Shimla	JFICT14	13 October 2026	_____

~ END OF REPORT ~